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A study of a bluff-body combustor using laser sheet lighting *

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Abstract. Laser sheet lighting is used to study reacting flows with and without heat release in an axisymmetric, unducted and vertically mounted bluff-body combustor. The fuel, which is seeded with titanium tetrachloride vapor, is ejected from a jet located in the center of the bluff-body. The $TiCl_4$ in the dry fuel reacts spontaneously with the water in the annulus air to form titanium dioxide particles. High speed movies and visual observations of vertically and horizontally located sheets of laser light provided remarkably detailed visualization (via Mie scattering) of the vortex dynamics in the near-wake region of the bluff-body.

1 Introduction

The Aero Propulsion Laboratory (APL) has developed a bluff body research combustor for the purposes of: (1) evaluating the performance of conventional probes and advanced laser techniques and (2) conducting experiments to aid in evaluating and developing combustion models. Numerous experiments and theoretical analyses have yielded a detailed understanding of many of the characteristics of this combustor (see Roquemore et al. 1983). High speed movies have proven to be the most useful technique for understanding the turbulent combusting flow field. They revealed the existence of large scale vortices shed from the bluff body. These vortices are an important part of the mixing process in the near and far wake regions. The interpretation of single point Laser Doppler Anemometer (LDA) and Coherent Anti-Stokes Raman Spectroscopy (CARS) measurements in terms of the large scale structures has been greatly enhanced by the use of high speed cine pictures. Flow visualization coupled with simultaneous single or multiple point measurements is believed to be essential for understanding the details of flows containing large scale structures.

Sheet-lighting has been used in recent years to provide quantitative two-dimensional (2-D) visualization of flow fields. Long et al. (1981) used a Lorenz/Mie scattering technique in a aerosol seeded jet to study the entrainment and mixing processes. Escoda and Long (1983), in a similar study, used Rayleigh scattering to study a freon jet. Kychakoff et al. (1982) used fluorescence scattering to record the 2-D view of the OH radical formed in a flame. Johnson and Bennett (1981) studied turbulent mixing processes of coaxial jets discharging into an expanded duct using a fluorescent tracer in a water tunnel experiment. Koochesfahani et al. (1983) and Dimotakis et al. (1983) also used a fluorescent dye in water tunnel experiments to study the mixing in chemically reacting shear layers and jets, respectively. These studies have demonstrated the value of different sheet-lighting techniques for studying flows containing large scale structures.

The first objective of this paper is to introduce a two-dimensional (2-D) sheet-lighting technique coupled with a fast chemically reacting system that can provide detailed visualization of turbulent mixing processes without heat release. This technique also offers the potential for making simultaneous velocity and relative product concentration measurements. Visualization results obtained in a small unducted bluff-body combustor, of similar design to that used by Roquemore et al. (1983), are presented as a demonstration of the technique. Micron size TiO_2 particles are used as the scattering media. These particles are formed as a product of the reaction of $TiCl_4$ and H_2O. This reaction has been used by several investigators to provide seed particles for LDA measurements (Moss 1980; Craig et al. 1984). To our knowledge, however, the potential use of this reactive system as a tool for studying turbulent mixing processes has not been explored. The TiO_2 particle size has been reported to be fairly uniform and is generally in the 0.2 to 1 micron range with a small fraction in the 2 to 3 micron range (see Craig et al. 1984). Particles in this size range can follow the flow and also provide good Mie scattering centers. Normally, large concentrations of TiO_2 particles are formed so that the prospects of obtaining unbiased LDA results by uniformly sampling, as demonstrated by Craig et al. (1984), appear good. Also, the work of Starner and Bilger (1980) suggests that the

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scattered light intensity from a measurement volume in a non-reacting flow might be directly proportional to the TiO_2 concentration. If so, this technique might also provide 2-D quantitative data on product concentrations in cold flows; however, this point needs to be carefully examined before such a claim can be made. This technique cannot be used for quantitative measurements in combusting flows because the scattering cross section of TiO_2 particles is greatly reduced in the flame (Ebrahimi & Kline 1977).

The $TiCl_4 + H_2O$ reaction seems to be nearly instantaneous since TiO_2 particles begin to form just at the exit of a high speed jet in which a uniform mixture of $TiCl_4$ and dry air is injected into humid air. This indicates that the reaction is mixing limited and can be effectively used to study turbulent mixing processes. The corrosive nature of $TiCl_4$ and HCl may have discouraged its use in the past but with proper handling procedures and good ventilation and exhaust systems, this does not appear to be a serious problem.

The second objective of this paper is to use the 2-D sheet-lighting technique to investigate the role of vortex shedding in the bluff body combustor. In reacting flow experiments without heat release, snapshots and high speed cine pictures of the near wake region behind the bluff body clearly show that large scale vortices are shed from both the central jet and the bluff body. Much smaller vortices are formed inside and outside of the larger vortices which gives the flow a 3-D character. Under appropriate flow conditions, these vortices can interact in spectacular ways which are clearly evident in the high speed movies but are less discernible when shown by the individual frames. Sheet-lighting is used to qualitatively examine the roles of these vortices under different operating conditions. Vortex shedding from the outer edge of the shear layer of the bluff body did not occur for the combustion flow conditions. However, vortex shedding from the central jet was observed and its role in the mixing process is discussed. These observations were made possible by adding $TiCl_4$ vapor to the propane and using the sheet-lighting to illuminate the soot and TiO_2 particles formed in the combustion process.

2 Experimental set-up

A schematic diagram of the experimental set-up is shown in Fig. 1. The bluff body combustor was symmetrically mounted in a small vertical combustion tunnel. The fuel nozzle located at the center of the 60 mm diameter bluff body is 4.78 mm in diameter. This nozzle has a short-tapered contour as recommended by ASME for providing a flow with a flat velocity profile and low turbulence intensity at the exit. The combustion tunnel was also designed to provide low turbulence intensity and uniform velocity air to the test section. The tunnel, with an 80 mm diameter, can be operated either ducted or unducted. For the experiments reported in this paper, it was unducted and thus subject to room air currents. During the experiments, the only ventilation system operating was the exhaust hood which was placed about 50 cm directly above the tunnel. Though precautions were taken to still the room air, disturbances in the combustor flow field, attributed to room air currents, were occasionally observed, especially when the combustor was operated at annulus velocities of 2 m/s or less.

The 60 mm diameter bluff body has also been operated in the 254 mm diameter, ducted combustion tunnel at APL. Although the results from these experiments will not be presented in this paper, several observations are noteworthy. In the large ducted tunnel, the flame was attached to the bluff body face at annulus velocities in excess of 20 m/s; whereas in the small unducted tunnel, the flame lifted from the face when the velocity exceeded about 4 m/s. If the small tunnel was operated long enough for a layer of soot about 2 mm thick to form on the bluff body face, the flame would remain attached for velocities up to 8 m/s. For the high annulus velocity combusting experiments reported in this paper, the soot layer was used to stabilize the flame. About a $\pm 2.5\%$ variation in the annulus velocity at different spatial locations was noted in the small tunnel. This variation did cause some asymmetry in the flow field which could have contributed to the flame detachment problem. Vortex shedding from the bluff body was observed in the large tunnel with an annulus velocity of about 18 m/s but not in the small tunnel where the velocity was about 8 m/s. Future studies will examine the effects of velocity, blockage ratio, ducted vs. unducted test section and heat release on the vortex shedding process.

The central jet flow rate was measured with calibrated rotameters. A pressure regulator was used to set and main-

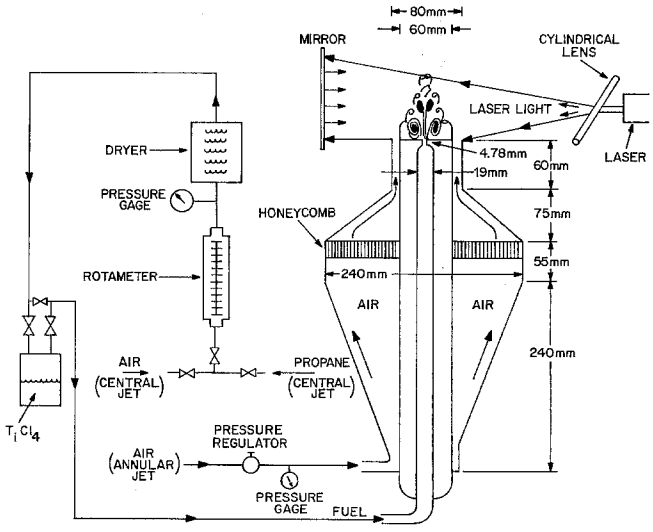

Fig. 1. Illustration of the experimental set-up

tain a given flow rate of air to the annular region of the nozzle. A pitot tube and precision manometer were used to determine the velocity at the annulus exit. This determination was made prior to each test run.

Experiments were conducted with shop air flowing in the annular region and either dry air or propane, containing small amounts of $TiCl_4$ vapor, flowing through the central jet. Some experiments were conducted to show the differences in the flow field with and without heat release. In these experiments, $TiCl_4$ vapor was added to the propane but not ignited. In some combusting experiments, $TiCl_4$ vapor was added to the propane to increase the particle concentration so that cine pictures could be made.

The optical set-up consists of a 4-watt Argon Ion laser, which is operated at the 5,145 Å Argon ion line. The laser beam strikes an 8 mm diameter quartz rod which acts as a cylindrical lens. The light beam fans out in a plane perpendicular to the axis of the quartz rod. The angle of divergence depends on the rod diameter (greater for smaller diameters). The thickness of the light-sheet is approximately equal to the diameter of the laser beam exiting from the laser (1.25 mm). To increase the intensity of the sheet-lighting of the test section, a plane front-surfaced mirror was placed behind the test section. The position of this mirror was adjusted so that the light-sheet is reflected back on itself. In this manner, the light intensity at the test section is increased nearly twofold. Tests were also conducted with horizontal and vertical sheets of light simultaneously appearing in the test section. To accomplish this, a beam splitter, another cylindrical rod, and a mirror were added to the optical set-up. A variable speed chopper was installed at the laser output to allow strobing of the light sheet. This made it possible to observe, with the eye, the dominant frequencies of the vortices and their length scales.

Because of the continuum radiation from hot carbon particles in the propane flame, the laser light scattered by the cooler carbon and TiO_2 particles within certain sections of the flame zone was difficult to observe. This difficulty was overcome by using a narrow, band-pass optical filter. The filter used was centered at 5,145 Å and had a half-width of 90 Å. The light intensity at 5,145 Å was reduced 40%. With this filter in place, the yellow continuum radiation from the carbon in the reaction zone was nearly eliminated; thus, allowing one to observe the sheet-lit plane within the propane flame.

Still pictures (35 mm) and high speed motion pictures were taken of the sheet-lit reacting flow. At the beginning of each reel (or roll) of film, a target (10 mm grid) was mounted over the center of the nozzle, where the laser sheet was located. The target was used to focus the camera as well as provide a grid for making measurements from the photographs. The sheet-light intensity was sufficient to allow still pictures, in most instances, to be taken at f4.0, 1/2,000 second. Motion pictures were taken, in most instances, at f2.8 and 1,000 frames per second. The shutter speed of the motion pictures is approximately 1/2,600 of a second. To take pictures when the sheet-light is horizontal, a mirror was installed at about 60 cm above the nozzle. Thus, it was possible to make photographs directly above the nozzle with the optical axis of the camera perpendicular to the horizontal sheet-light. When the optical filter was in place, it was necessary to increase the exposure time one stop.

Tests were conducted over a wide range of operating conditions. For example, in the cold flow experiments, the Reynolds number of the annular flow varied from 1,250 to 22,500, the central jet from 1,250 to 22,500. For the propane experiments, the Reynolds number of the annular flow, calculated from the inlet conditions, varied from 2,500 to 10,000; the central jet from 500 to 11,000.

3 Results and discussions

This section is organized into two subsections: Reacting Cold Flow and Combusting Flow. The results obtained from observing the mixing between $TiCl_4$ vapor plus dry air ejected from the central jet and moist air ejected from the annulus are presented in the reacting cold flow subsection. Also, the locations of the forward and aft stagnation points for a wide range of annulus velocities are presented. Descriptions of the flow field that result for different fuel velocities and a trace of the time history of shed vortices are also presented. The combusting flow fields are described in the combusting flow subsection. $TiCl_4$ vapor was added to the central propane jet to aid in visualizing the flow field through the flame. Also, some photographs of the non-combusting propane and $TiCl_4$ mixture are presented.

3.1 Reaction cold flow

3.1.1 Location of forward and aft stagnation points

The bluff body combustor can be thought of as two widely spaced jets, an annular jet and a central fuel jet. A recirculation zone is established as a result of the expansion and entrainment of the annular jet. The reverse flowing air in the recirculation zone is in direct opposition to the flow of the central jet. For flow conditions at which the momentum of the reverse flowing air is larger than the momentum of the central jet, the central jet flow will be turned back towards the bluff body. The axial velocity of the central jet is zero at the point where this occurs. The centerline location, where the central jet is turned, is called the forward stagnation point. The maximum length of the recirculation zone along the centerline is called the aft stagnation point. When the momentum of the central jet is larger than that of the reverse flowing air, the central jet will penetrate the recirculation zone, and hence no stagnation points exist along the centerline (Fig. 3).

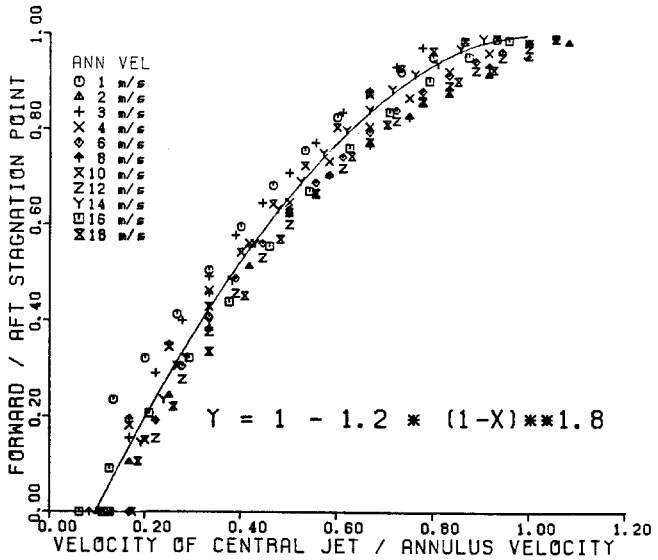

Fig. 2. Location of the forward stagnation point for different velocity ratios

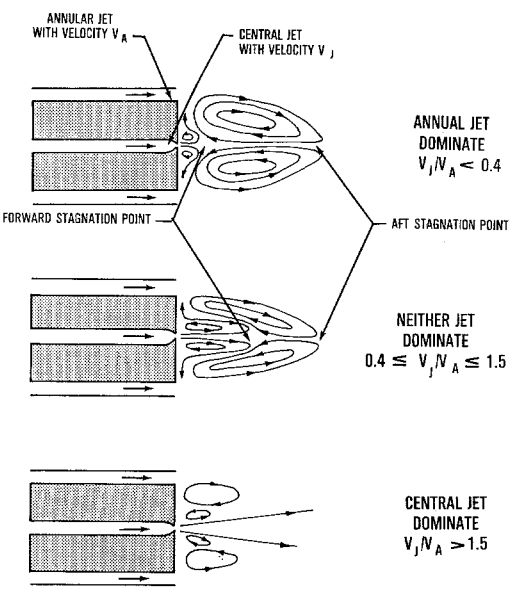

Fig. 3. Illustrations of the time-averaged flow fields for different air and fuel flow rates

The forward and aft stagnation points were measured over a wide range of annular and central jet velocities. These measurements were made by simply mounting an *L*-shaped tube on a vertical traversing mechanism and using it as a positioning probe that could be readily moved in and out of the test section. Some judgement is required on the mean location of the stagnation points because of fluctuations, but by visually averaging, the results appear to repeat fairly well.

The procedure is as follows: The annular air flow is set at a prescribed flow (determined by pitot measurements) and the central fuel jet ($TiCl_4$ vapor and air) is varied incrementally until the jet penetrates. At each incremental change, the position of the forward stagnation point is measured. Two sets of data were collected on different days. Except for the lowest annular flow rates (1 to 3 m/s) the variation between the data points was generally less than 10%. A total of approximately 500 measurements were made. The data for each annulus velocity were averaged and a smooth curve was drawn through the points.

The forward stagnation points normalized to the aft stagnation point for different central to annulus jet velocity ratios are shown in Fig. 2. The data points were obtained from smooth curve fits of the averaged data as previously explained. The aft stagnation point was found to have a constant value of 49 mm for annulus velocities greater than 4 m/s. Using the measured aft stagnation point as the normalizing parameter, all the data collapse to a single curve. The equation for the curve is given on the graph. The observation that penetration occurs at a velocity ratio of one should not be generalized since a different size bluff body or central jet would very likely give a different result. These data suggest that the time average flow fields are similar for conditions where the ratio of annular velocity (V_A) to central jet velocity (V_J) is a constant.

3.1.2 Flow field description

Of necessity, turbulent recirculating flows are usually described in terms of time averaged flow fields. Visual observations of the sheet-lit reacting cold flow field in the wake of the bluff body, tend to provide a time averaged view. Viewed by eye, flow conditions with the same velocity ratio (V_J/V_A) appear similar. In general, one observes a large recirculation zone (a toroidal vortex) which is bound to the bluff body and is driven by the annular jet. The central jet flow extends into the center of the recirculation zone. A second recirculation zone, which is driven by the central jet as it entraines fluid, appears narrow and is stretched along the stem of the central jet. In an attempt to characterize the visually observed flow fields, three general flow conditions were identified. They are consistent with those described by Roquemore et al. (1983) and are illustrated in Fig. 3. The first condition is characterized by the domination of the annular jet. The second is where neither jet dominates and the third is where the central jet dominates. Each of these will be discussed in the following paragraphs.

The annular jet appears to dominate the flow field for small velocity ratios (V_J/V_A) up to a value of about 0.4. For these conditions, the momentum of the reverse flow is sufficiently large to turn the central jet flow before it reaches the height of the time averaged center of the large recirculation zone. The vortex center is approximately 25 mm above the bluff body. In this case, the central jet is

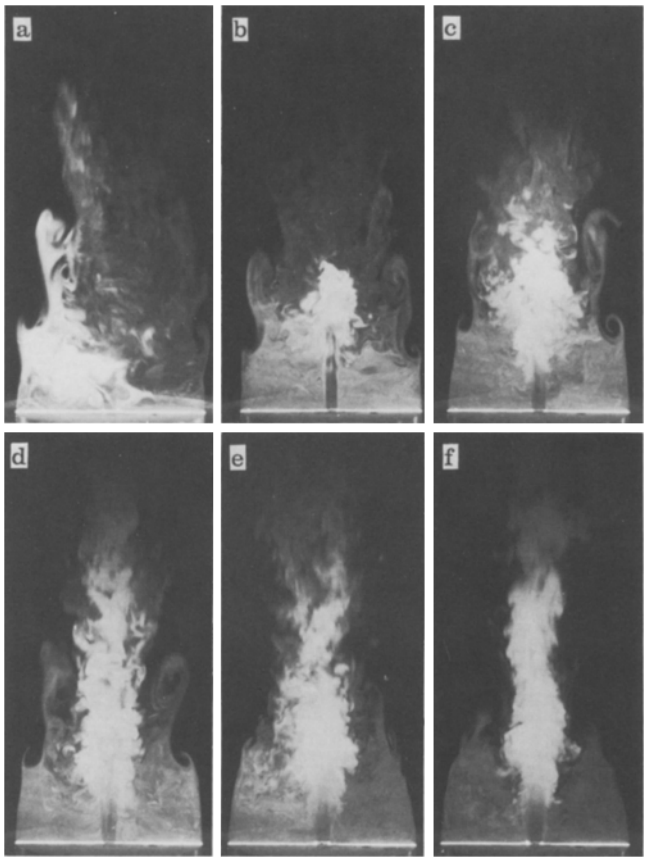

Fig. 4. a – f. Photographs for a fixed annulus velocity of 4 m/s and central jet ($TiCl_4$ vapor and dry air) velocities of: **a** 1, **b** 3, **c** 6, **d** 8, **e** 12, and **f** 15 m/s

very unstable and moves from side to side in an erratic way. Figure 4a shows a snapshot of the flow field with the annular jet dominant. Although the snapshot does not represent the time averaged view shown in Fig. 3, it does clearly show the asymmetry of the flow field produced by the central jet being pushed to the left at the instance the picture was taken.

Neither jet appears to dominate the flow field for velocity ratios between 0.4 and about 1.5. When the momentum of the central jet is sufficient to move the forward stagnation point above the axial height of the large vortex center, the stem of the jet does not move. However, the tip of the jet does fluctuate a few millimeters in all directions. As the velocity ratio approaches one, the central jet penetrates about 50% of the time and at a ratio of about 1.5, it penetrates 100% of the time. Snapshots of the flow field, where neither jet dominates, are shown in Figs. 4b and 4c.

The central jet appears to dominate the flow field for velocity ratios greater than 1.5. Snapshots of this case are shown in Figs. 4d – f. The central jet appears to entrain a large quantity of annulus air. The annulus air is pulled in around the shear layer of the annular jet and back along the shear layer of the central jet. This air path is clearly shown by the dark regions in Fig. 4f. A large central jet entrainment rate will also reduce the size of the recirculation zone.

Although the velocity ratio is a good parameter to characterize the time averaged flow field, it is not sufficient. Flow fields at low and high velocities, but with a constant velocity ratio, appear very different when the laser sheet is chopped. Large spiral vortices are apparent in the shear layer of the annular jet and other vortices are observed in the central jet. Some of these vortices are captured in the photographs in Fig. 4. These vortices can interact in numerous ways depending on the Reynolds numbers of the two jets. For annulus velocities below 1 m/s the recirculation zone appears to be laminar. At velocities between 1 and 8 m/s, quasi periodic spiral vortices are formed in the shear layer of the annular jet. The size of the spiral vortices is reduced as the velocity is increased. At 8 m/s a notable change in the noise level occurs and high frequency random vortex shedding is apparent. We believe the flow transitions to fully developed turbulence at this point. A similar transition from laminar to turbulent flow was noted in the central jet. It appeared laminar for velocities below 1 m/s and transitioned to turbulent flow at about 4 m/s.

The photographs in Fig. 5 are from a high speed movie of the flow field with an annulus velocity of 4 m/s and a central jet velocity of 4 m/s. Numerous details show up in the high speed movie that are not clear in the photographs. However, several points are evident from Fig. 5. At the beginning of the time sequence, the spiral vortices are not shed symmetrically, but towards the end of the sequence symmetrically shed vortices are noted. In the beginning of the sequence, the two small vortices on the right go through two pairing processes to form a rather large vortex that is mixed into the downstream wake of the bluff body. In this case, the vortex transported fluid out of the recirculation zone; it also contributed to the fuel and air mixing process. More study will be required to determine whether a significant amount of product, the observed particles, is formed in the large vortices as noted in the reacting shear layer experiments of Breidenthal (1981), or whether most of the product is formed before being rolled up in the vortices.

Figure 6 shows the dynamic events occurring in a horizontal plane 37 mm above the face. In the movies, a radial oscillation is seen where the flow expands and contracts in roughly a ring shape. One cycle of the expansion and contraction corresponds to a large spiral vortex, as noted in the vertical plane, passing through the observation plane. In reality, the vortex has a toroidal shape but, within the toroid, the flow is definitely three-dimensional and clustering of small horizontally oriented vortices can be observed. Much of the complex structure of the toroid is due to the different ways the spiral vortices interact as noted in the discussions of Fig. 5. The

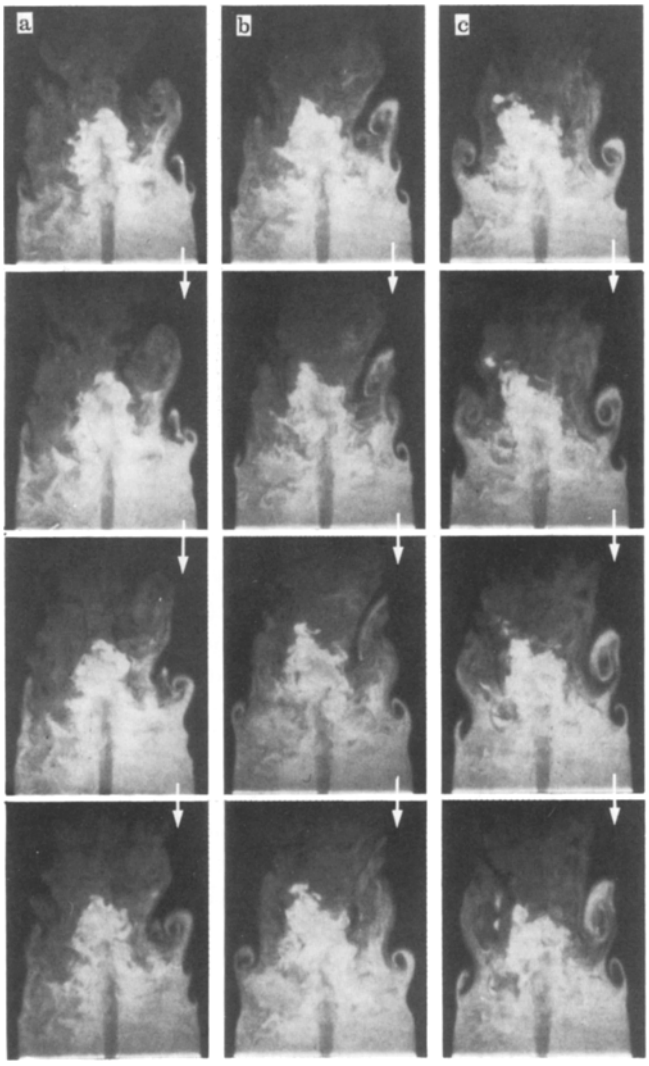

Fig. 5a–c. High speed cine pictures (1,000 frames/s, every other frame shown) for an annulus velocity of 4 m/s and a central jet ($TiCl_4$ vapor and dry air) velocity of 4 m/s

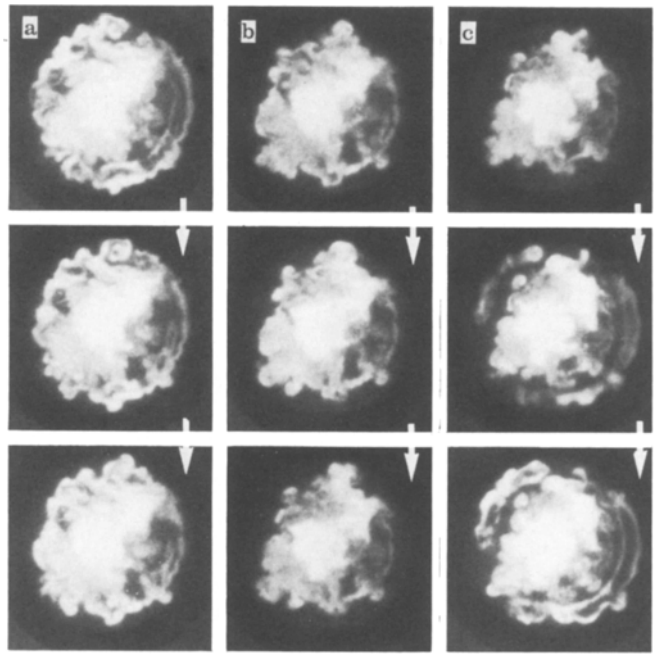

Fig. 6a–c. High speed cine pictures (1,000 frames/s) of a sheet-lit horizontal plane located 37 mm above the bluff body for an annulus velocity of 4 m/s and a central jet ($TiCl_4$ vapor and dry air) velocity of 4 m/s

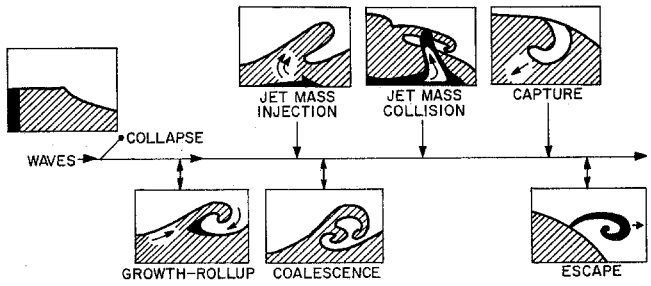

Fig. 7. Illustration of the time history of vortices shed from the bluff body

dark areas shown in the photographs in Fig. 6 represent air that has been entrained by the spiral vortices. Considerable interaction between the vortices in the ring and the central jet is also occurring at this horizontal location.

3.1.3 Time history of vortices shed form the bluff body

The development of a toroidal (spiral) vortex in the shear layer of the annular jet depends on how it interacts with the central jet and other vortices as it is convected downstream. Typical events (deduced from the study of high speed movies) are shown schematically in Fig. 7. The vortices begin as waves on the shear layer of the annular jet. Occasionally, these waves are observed to collapse before significant growth occurs. This failure to grow and mature is apparently caused by a reduction in the velocity differential across the shear layer to the point where exponential growth does not occur. The movies indicate that this happens when, by chance, recirculating mass is projected along the inside surface of the shear layer with approximately the same velocity and in the same direction (no appreciable radial velocity component) as the annulus air. Normally, however, the interfacial waves grow and roll up into a characteristic spiral figure. Rapid growth occurs just upstream of the axial location of the time-averaged vortex center of the recirculation zone. This rapid growth is attributed to a high probability that fluid elements, which collide with the shear layer, will have a large positive radial velocity component due to the turning of the flow around the vortex center. This collision often causes the surface waves on the shear layer to

undergo a large outward radial displacement that assists in the exponential growth of the wave into a toroidal vortex.

The toroidal vortex is frequently distorted or broken up by interaction with mass shed from the central jet. Jet mass (with a large radial velocity component) streaming into a spiral arm may cause the arm to swell and cease rolling up. Mass shed from the central jet, especially near the jet stagnation point in nonpenetrating cases, often has a strong radial velocity component. In such cases the jet mass collides with vortices moving axially on the recirculation zone boundary, usually destroying the spiral structure.

Vortices also interact with each other as noted in Fig. 5. If the time interval between the formation of two vortices is short enough, the second one will roll up through the base of the first as they move along the recirculation zone boundary. The first vortex may be cut in two as the second one rolls through it. Part of the cut vortex is pushed back into the recirculation zone and part rolls up with the second to become a coalesced pair which continues rolling about the common center as it is convected downstream. As previously noted, more than one paring may occur during the life of the vortices.

As vortices approach the tip of the recirculation zone, they usually break up and part of their mass escapes and is convected downstream while the other part is captured by the reverse current near the tip and drawn back into the recirculation zone. This mechanism is postulated to be a primary means of mass exchange between the recirculation zone and its surroundings when vortex shedding is present. As spiral arms of recirculation zone fluid stretch and roll up, they normally engulf a large excess of fluid from the annular air stream. When a vortex is captured, either in whole or in part, near the tip of the recirculation zone, this engulfed air is often transported, virtually unmixed, deep into the recirculation zone. Thus, a primary entrainment mechanism is governed by the dynamics of large scale vortices shed from the bluff body. A similar observation has often been noted in studies of 2-D shear layers (see Dimotakis et al. 1983; and Breidenthal 1981).

3.2 Combustion flow

3.2.1 Flow field description

The first photograph in Fig. 8 shows a flame stabilized on the bluff body face where the annulus velocity is 2 m/s and the central propane jet velocity is 0.5 m/s. The second photograph shows a laser sheet-lit view of the same flame taken with an optical filter to eliminate light from sources other than the laser. The Mie scattered light from the soot and TiO_2 particles clearly show the laminar structure of the flame. Surrounding the central jet is a bound toroidal vortex. The vortex center is located approximately 20% of the bluff body diameter above the base. The dark region on the right-hand side between jet surface and vortex ring shows an air entrainment pathway into the recirculation zone. In high speed movies, such entrainment pathways are observed to exist for extended periods of time compared to the period of rotation of the vortex. In a movie showing this flame from above, a pair of these pathways is frequently seen, spaced from 120 to 180 degrees apart. These results indicate that an axisymmetric model is inadequate for this flame. The central jet also has a large mushroom shaped vortex near its tip where the jet is turned by the reverse flowing air of the recirculation zone. The growing vortex on the jet is eventually shed, stretched, and rolled up in the recirculating flow.

The third photograph in Fig. 8 shows a sheet-lit view of non-combusting flow with propane and $TiCl_4$ in the central jet and the same inlet conditions as the combusting case. In contrast to the exothermic flow, the cold flow is in a transition state with active vortex shedding from the bluff body. Comparison of the cold flow structure with that of the hot flow confirms that the latter flow is strongly affected by combustion laminarization due to the large increase in kinematic viscosity at higher temperatures (approximately a factor of 8). A similar difference between cold and combusting jets has been noted by Yule et al. (1980).

Figure 9 shows a propane flame with a central jet velocity of 4 m/s and an annulus air velocity of 1 m/s. With this increase in velocities, the flame surface at the mixing layer between the recirculation zone and the annulus air exhibits a train of large amplitude interfacial waves. These waves are prematurely terminated at the tip of the recirculating wake before they mature into spiral vortices. The center photograph in Fig. 9 is a laser sheet-lit view which reveals the interior structure of this flame. The irregular arrangement of structures in the wake and upper portion of the central jet indicate that both the wake and jet are in a transition state. The third photograph shows the non-combusting case with the propane and $TiCl_4$ in the central jet. Comparison between com-

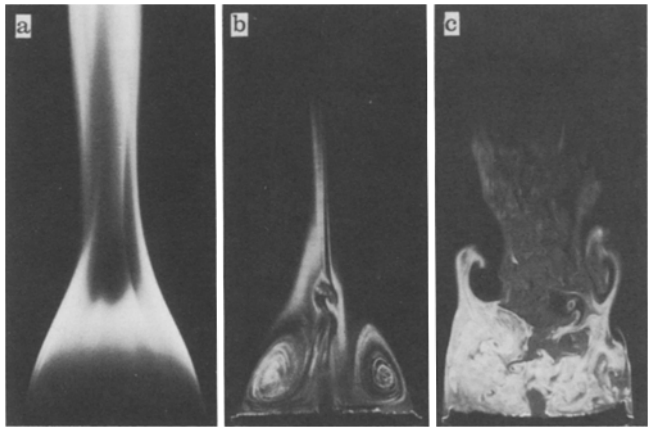

Fig. 8a–c. Photographs for an annulus velocity of 2 m/s and a central jet (propane + $TiCl_4$ vapor) velocity of 0.5 m/s. Left to right: the flame, sheet-lit flame, non-combusting flow

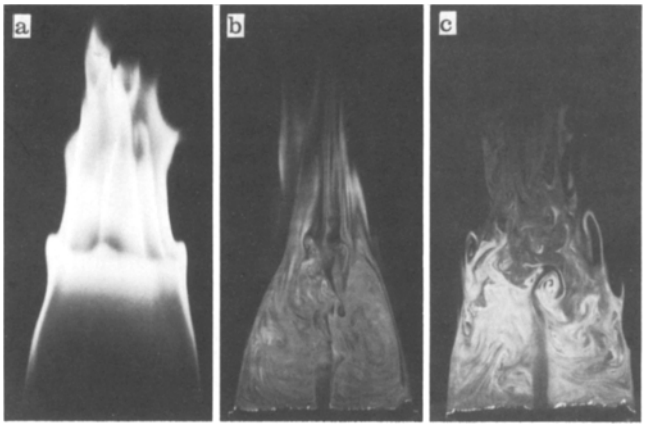

Fig. 9a – c. Photographs for an annulus velocity of 4 m/s and a central jet (propane mixed with $TiCl_4$ vapor) velocity of 1 m/s. Left to right: the flame, sheet-lit flame, sheet-lit non-combusting flow

busting and noncombusting cases again demonstrates the strong laminarization effect induced by combustion.

Figure 10 shows the 2-D evolution, in steps of three milliseconds, of a combusting flow with an annulus air velocity of 8 m/s and a central jet velocity of 10 m/s. The flow field is notably different than that shown in Fig. 8 where the annulus velocity is lower. The recirculation zone is believed to be in transition between laminar and turbulent flow. The central jet dominates the flow field and the entrainment of fluid near the base of the central jet appears to be responsible for the large quantity of reverse flowing air brought in around the aft end of the recirculation zone. The strong reverse flow of entrained air induces vortex shedding from the central jet and also causes vortices to form on the inner surface of the shear layer of the recirculation zone. These vortices can be seen to interact and pair in several places in the movie sequence. These interactions appear to be an important mixing mechanism in the recirculation zone. Spiral vortices are not formed in the outer shear layer of the recirculation zone as was observed in the cold experiments. Fujii and Eguchi (1981) also observed vortex shedding from a bluff body in cold flow but not in combusting flows.

The reverse flow of recirculating fluid and entrained air appears to induce shedding of large vortices from the central jet into the recirculating wake. Two major instances of this type of shedding can be seen in the right-hand side of the recirculation zone in the movie sequence of Fig. 10. In the second frame of the middle column, a large vortex extends towards the center of the recirculation zone from the right side of the central jet. This vortex interacts with a second vortex at the aft tip of the recirculation zone (last frame in the second column) to form a mushroom shaped vortex. This vortex appears to have a large radial velocity component. However, as it approaches the outer shear layer of the recirculating zone, the vortex

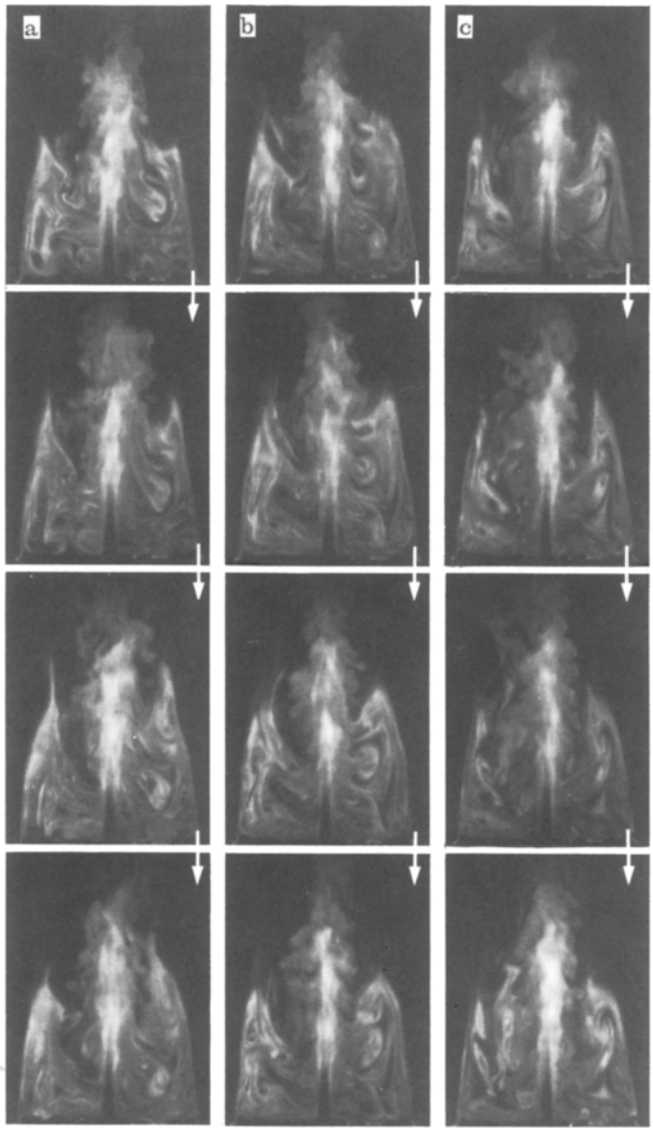

Fig. 10a – c. High speed cine picture of combustion (1,000 frame/s every third frame shown) for an annulus velocity of 8 m/s and a central jet (propane and $TiCl_4$ vapor) velocity of 10 m/s

is flattened and stretched axially. It does not extend into the shear layer. This process contrasts sharply with the cold flow case where mass, which is shed radially from the central jet, often moves out into the recirculation zone boundary where it deforms the vortices rolling up on that surface or creates new vortices.

4 Summary and conclusions

Laser sheet-lighting the product of a chemical reaction combined with photography, especially high-speed movies, is shown to be a very valuable tool in the study of a bluff body combustor. In contrast to seeding the upstream

flow with particles for visualization, sheet-lighting the product of a fast chemical reaction in the flow field provides a view of the mixing processes and how the product and air species are transported. The $TiCl_4$ water vapor reaction proved to be an excellent choice for use in this new program.

In this study, three primary modes of operation of the combustor with their characteristic flow structures are identified. These modes are a highly dominant annular jet mode, a highly dominant central jet mode, and an intermediate mode where the central jet penetrates the recirculation zone of the bluff body intermittently. Measurements of the central jet stagnation point normalized by the aft stagnation point of the recirculation zone are found to correlate with the ratio of central jet to annular jet velocity for the cold flow cases. Thus, the jet velocity ratio is found to be one of the parameters which can be used to characterize the time averaged flow field of an operating mode. The dynamics of large vortex structures appears to play a governing role in mass transport and mixing in all of the flows studied. Vortex shedding from the bluff body appears to be a primary mechanism of mass exchange between the recirculation zone, annular jet, and downstream region for the cold flow cases. The high speed movies reveal that the flow fields are not axisymmetric and steady, but rather, highly three-dimensional and unsteady in terms of the dynamic evolution of large scale vortical structures. In combusting flows, a strong laminarization effect due to heat release with consequent increase in kinematic viscosity was observed. This effect appears to prevent vortex shedding from the bluff body for the flow range of this study.

The success of this sheet-lighting visualization technique combined with high speed cine photography in this study opens up a wide range of possibilities for future studies. A few of these possibilities include: the study of the effect of blockage ratio on combustor operating modes and flow field dynamics, the study of ducted versus unducted flows, and a study correlating the visualization of vortex dynamics recorded in high speed movies with the results of simultaneous measurements made with advanced diagnostic techniques such as laser Doppler anemometer and coherent anti-Stokes Raman spectroscopy. The prospects of acoustically driving the flow and using a sheet-lit laser strobe for conditional sampling seems another attractive possibility as well as a means of studying instabilities in cold and combusting flows.

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